

Nicholas A. Anderson

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Education

PhD in Aerospace Engineering

University of Illinois at Urbana-Champaign

Advisors: Dr. Francesco Panerai and Dr. Gregory S. Elliott

August 2025

GPA: 3.49/4.00

Bachelors of Science in Aerospace Engineering

University of Illinois at Urbana-Champaign

May 2020

GPA: 3.54/4.00

Associates in Engineering Science

College of Lake County

May 2017

GPA: 3.61/4.00

Career Interests

Human space exploration; Hypersonics; Atmospheric entry, descent, and landing (EDL) systems; Plasma science

Experience

Engineering Research Intern

NASA Ames Research Center

Mountain View, CA

June 2024 - Aug 2024

- Material characterization of thermal protection system samples tested in high enthalpy arc jet facility using optical and scanning electron microscopy
- Developed program using Python and ImageJ to measure char layer depth and differential recession between regimes of reacted material
- Organized large set of experimental results into comprehensive narrative to be shared with research community

Engineering Research Intern

National Solar Thermal Test Facility, Sandia National Laboratories

Albuquerque, NM

May 2022 - Present

- Designed, constructed, and operated an atomic oxygen plasma chamber used in conjunction with the highly focused solar thermal heating of the Sandia Solar Furnace for materials testing
- Developed an experimental campaign with the ablation team to investigate surface interactions between atomic oxygen and carbon based heat shield materials
- Wrote pressure safety documentation for the new experimental facility and cooperated with pressure safety subject matter experts to compose operating procedure

Graduate Research Assistant

University of Illinois at Urbana-Champaign

Champaign, IL

Aug 2020 - Present

- Member of the Center for Exascale-enabled Scramjet Design (CEESD) research group, a part of the Predictive Science Academic Alliance Program (PSAAP)
- Modeling the oxidation processes of heat shield materials by measuring gas surface interactions and reaction rates between carbon and oxygen
- Experimentally investigating phenomena relevant to scramjet combustion environments to inform computational models and contribute to the readiness of these systems
- Designed and constructed a flow reactor of molecular and atomic oxygen with resistively heated graphite

Skills and Hobbies

Emergency Medical Technician

2025-Present

- Estimated certification May 2025; Basic life savings skills, e.g. CPR, airway management, trauma stabilization

General Aviation

2021-Present

- FAA licensed private pilot for fixed-wing, single engine, land aircraft under visual flight rules

Martial Arts

2002-Present

- Second Degree Black Belt in Taekwon Do
- Training in Hapkido, Brazilian Jiu-Jitsu, and western boxing

Publications

- Nicholas A. Anderson, Philip Zolfaghari, Souvik Bhattacharya, Lorenzo Capponi, Trey Oldham, R. Mohan Sankaran, Gregory S. Elliott, Francesco Panerai, Flow reactor experiments of high-temperature Graphite oxidation and nitridation, High Temperature Corrosion of Materials (2024).
- Nicholas A. Anderson, Lindsay Lawless, Lam Bahn, Kimberly D. Wakefield, Ricky Tang, Brian Bentz, Francesco Panerai, Jeffrey D. Engerer, Solar-Thermal Testing of Ablator Materials in an Atomic Oxygen Plasma, AIAA SCITECH Forum (2024)
- Nicholas A. Anderson, Souvik Bhattacharya, Kaan Kirmanoglu, Kelly A. Stephani, Gregory Elliott, R. Mohan Sankaran, Francesco Panerai, High temperature oxidation of graphite under non-thermal oxygen plasma, AIAA SCITECH Forum (2022)
- Kaan Kirmanoglu, Nicholas A. Anderson, Francesco Panerai, Kelly A. Stephani, Joseph C. Ferguson, Sigrid Close, Simulating Oxidation of Carbon Surfaces by Atomic Oxygen Coupled with a Finite Rate Oxidation Model, AIAA SCITECH Forum (2022)